

Sebastian Esteva

**CLASS PROJECT - 3-D PRINTING GLASSES
FRAMES**

Project 2 is what?

Design a part that includes at least one of the following features:

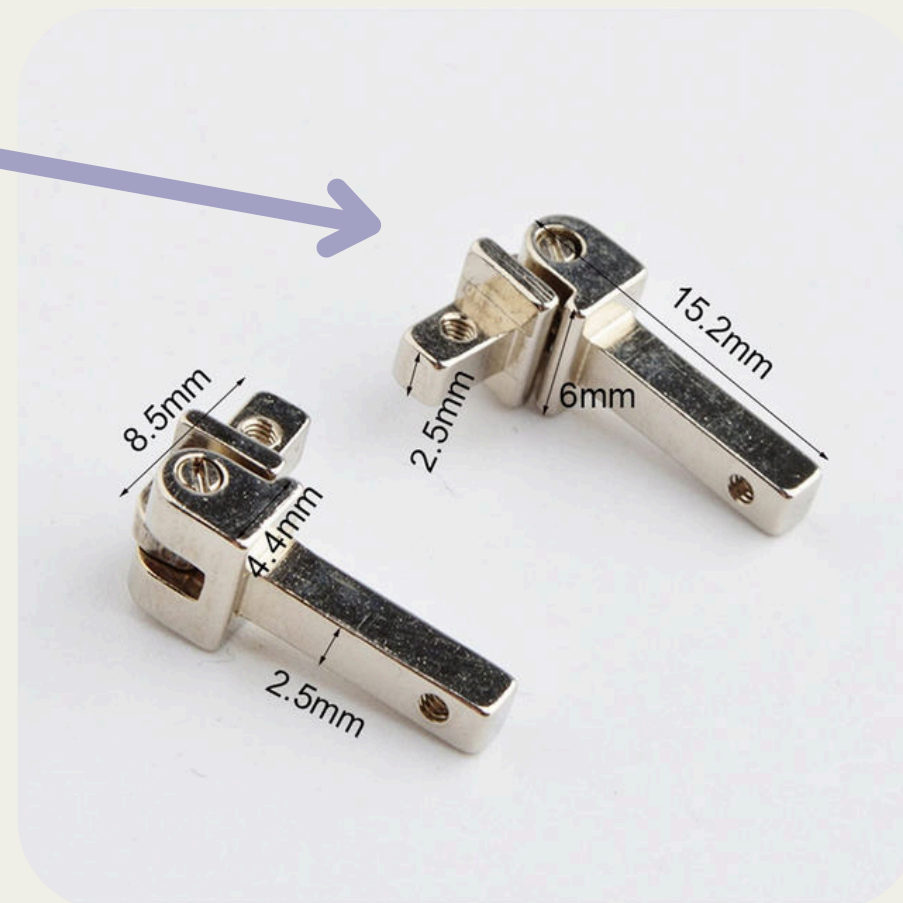
- Print-in-place
- Multiple materials
- Print-in-plate



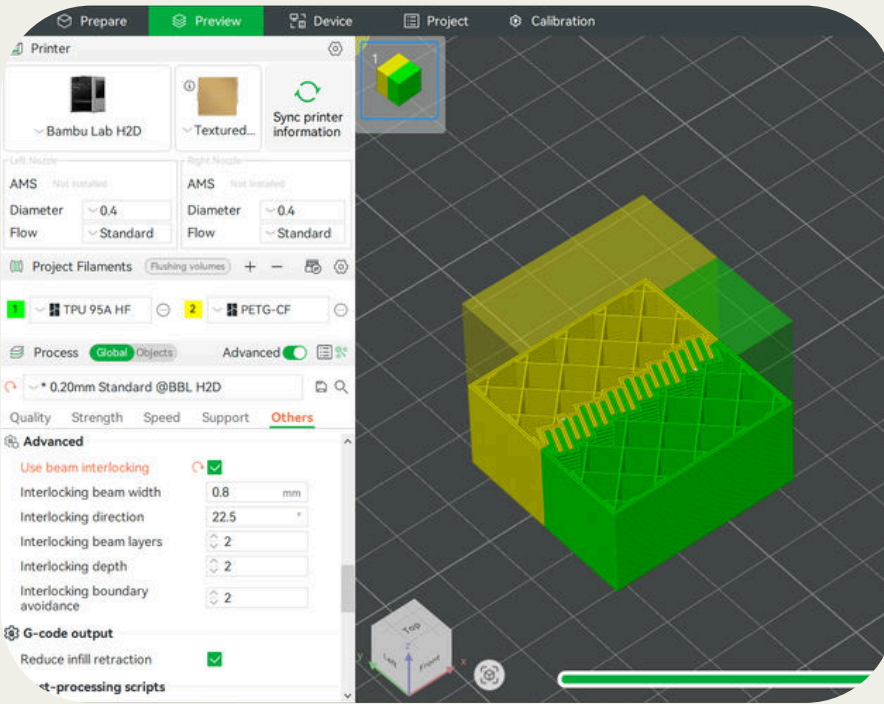
My Purpose:

Create a multi-material hinge for glasses.

Challenge:
Glasses Hinges
are small!
+
Aesthetics Are
Important



Multi-Materia Research



learnings

(1) Securely bond rigid and flexible materials using Beam Interlocking, (2) filaments need to dried.

Bambu.com



learnings

(1) Embraces the hinge as part of the design. (2) Use of flexible material for structural, and utility integrity.

"All You Need to Know..."



learnings

This print illustrated (1) the purposeful use of geometry for interlocking and (2) potential designs for a thin structure.

Multi-Material

Comparisons

Basic **Vs.** Multi Material 3D Printing



More materials =
more properties

One material = one set of properties.

Comparisons



4 pieces

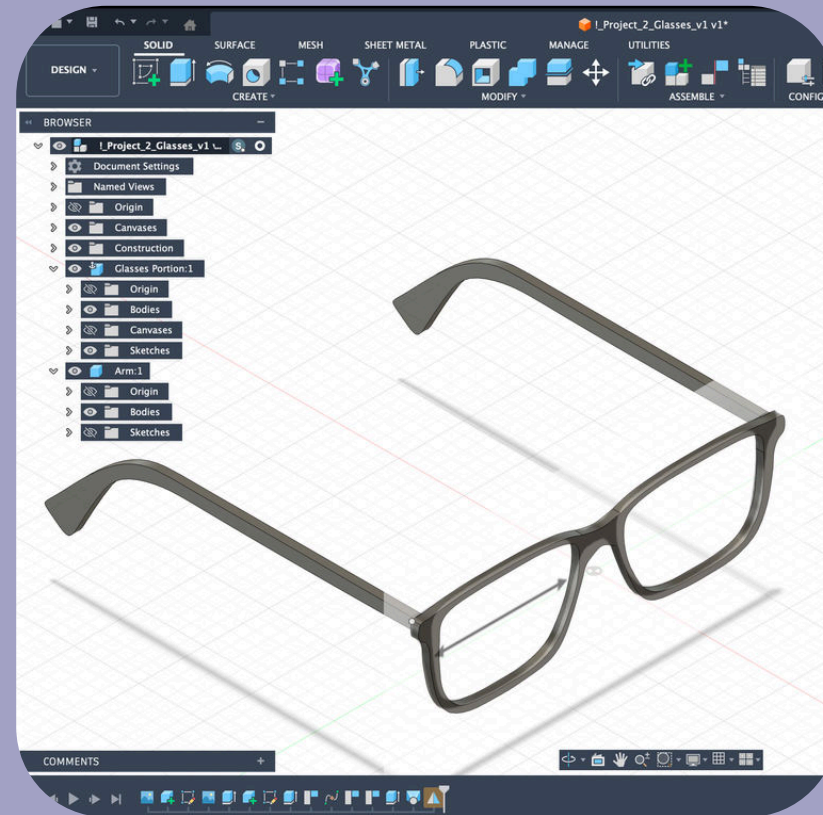
Traditional manufacturing vs. 3D Printing

	Traditional	3D Printing
Durable	+	(Can be with good design)
No Assembly		+
Few pieces		+

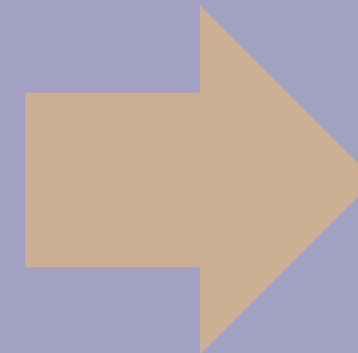
A Quick Aside: The *Frames* Journey



Wire frames
designed for a
perfect fit.

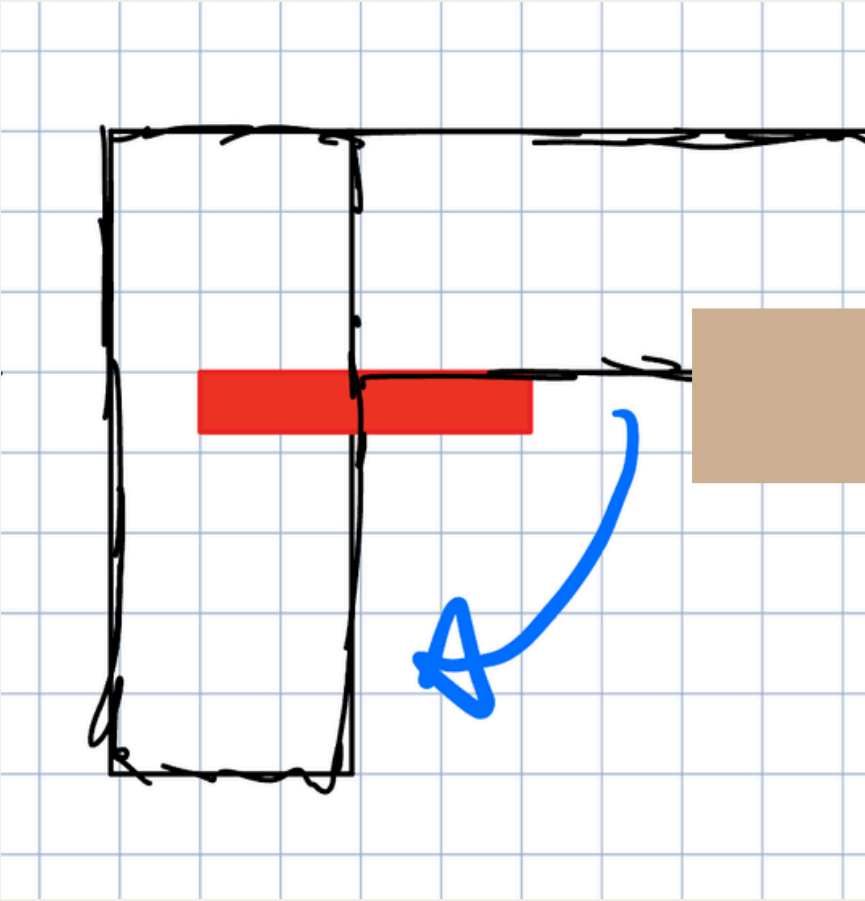


Measurements from
wireframes used for
design.

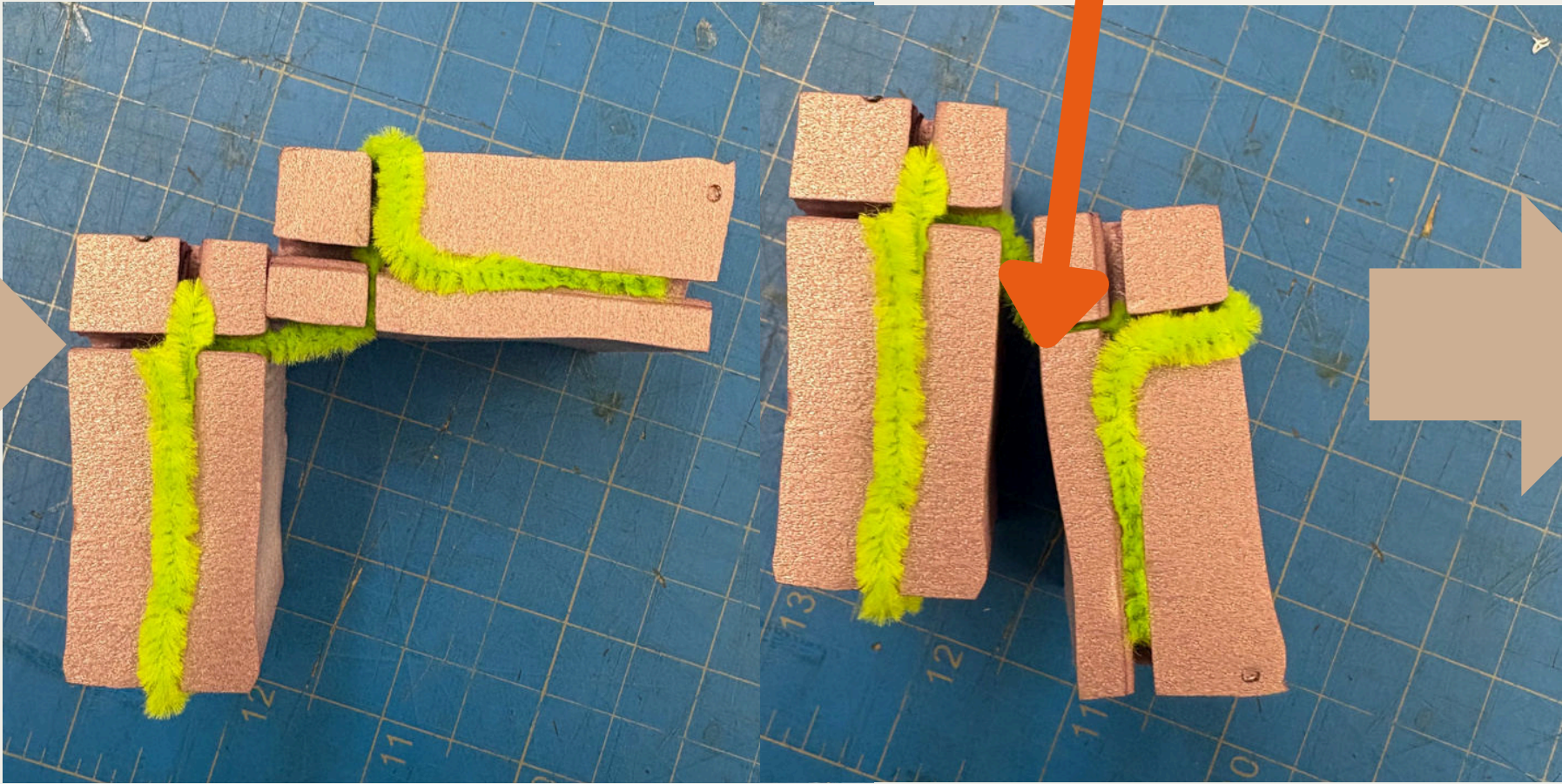


3D Printed
Prototype of Just
Frames

Idea Generation



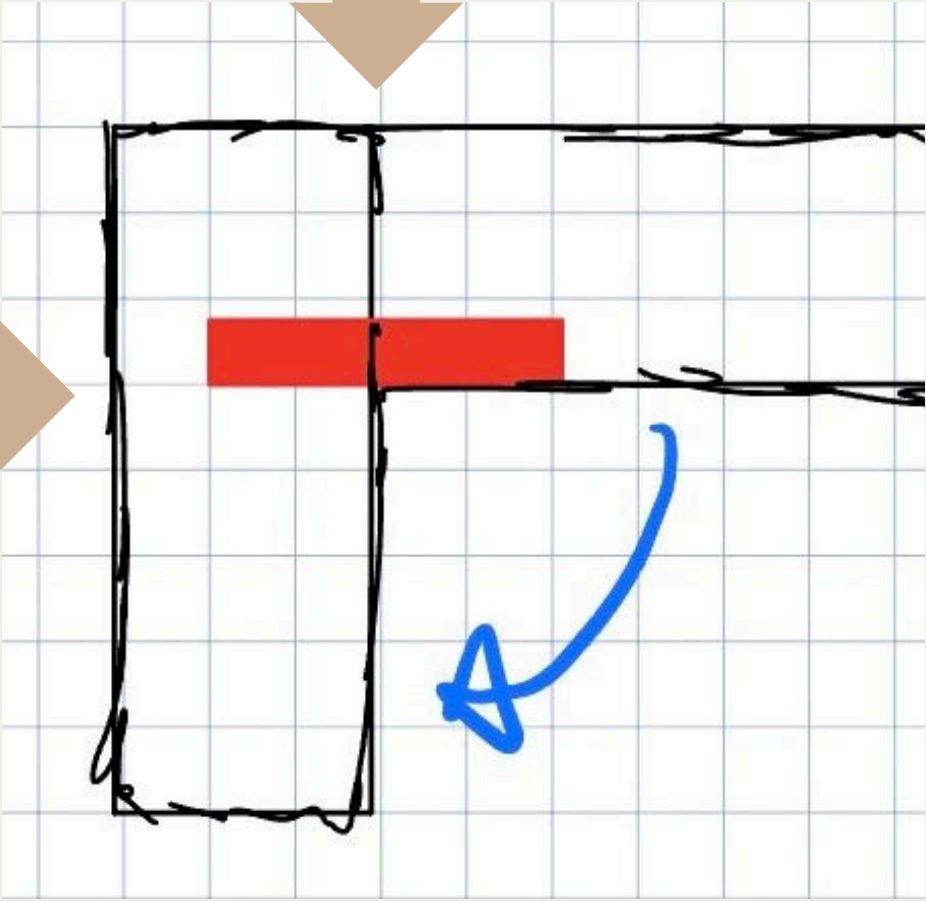
Base principles
simple, hinge



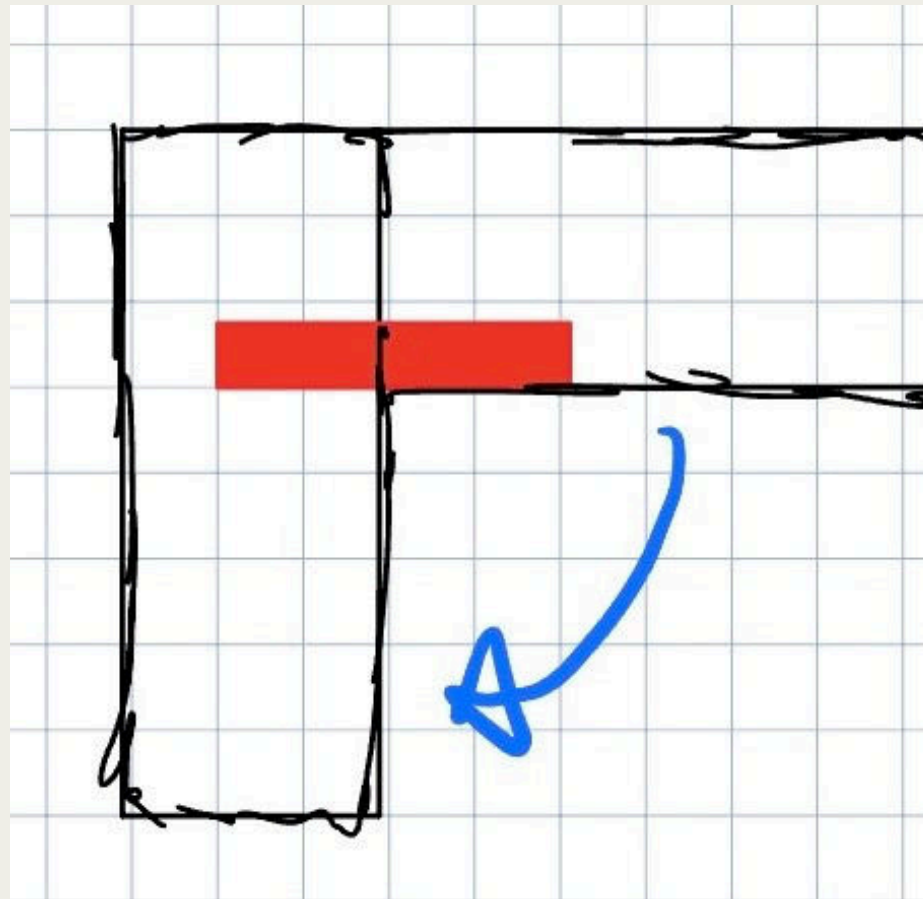
prototype

“Arm” portion
pushed out by
connector

moved connector in
for “flush” bend



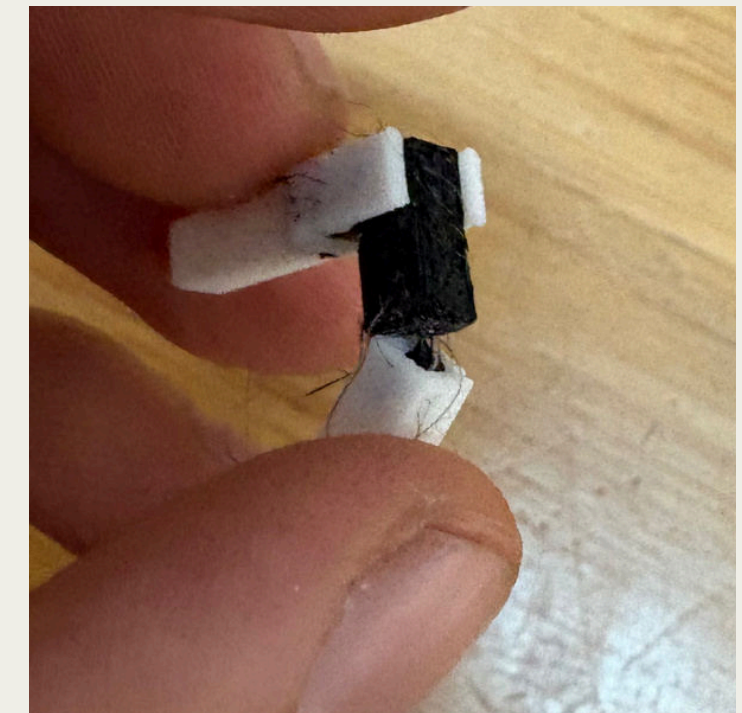
Idea Generation



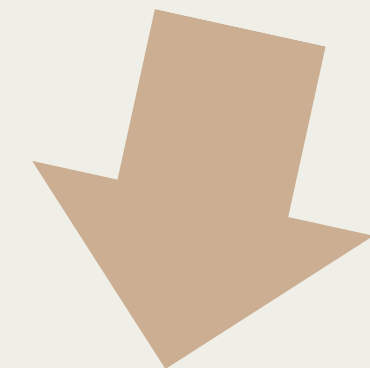
How do I bond
the two
materials?



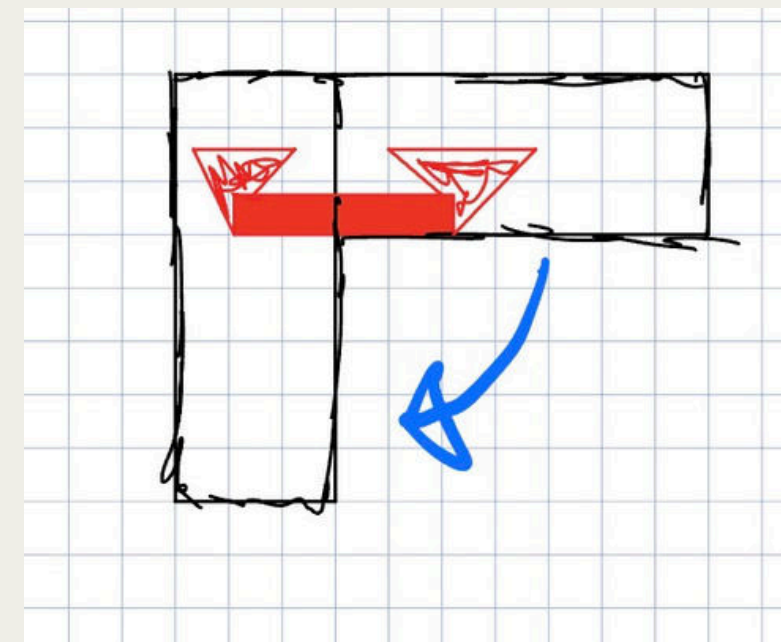
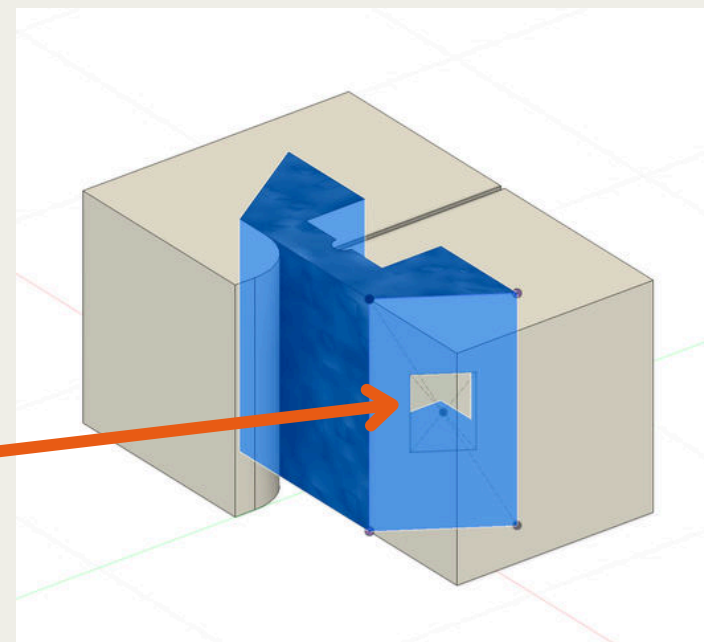
Cross
Bridges?



NO, too small

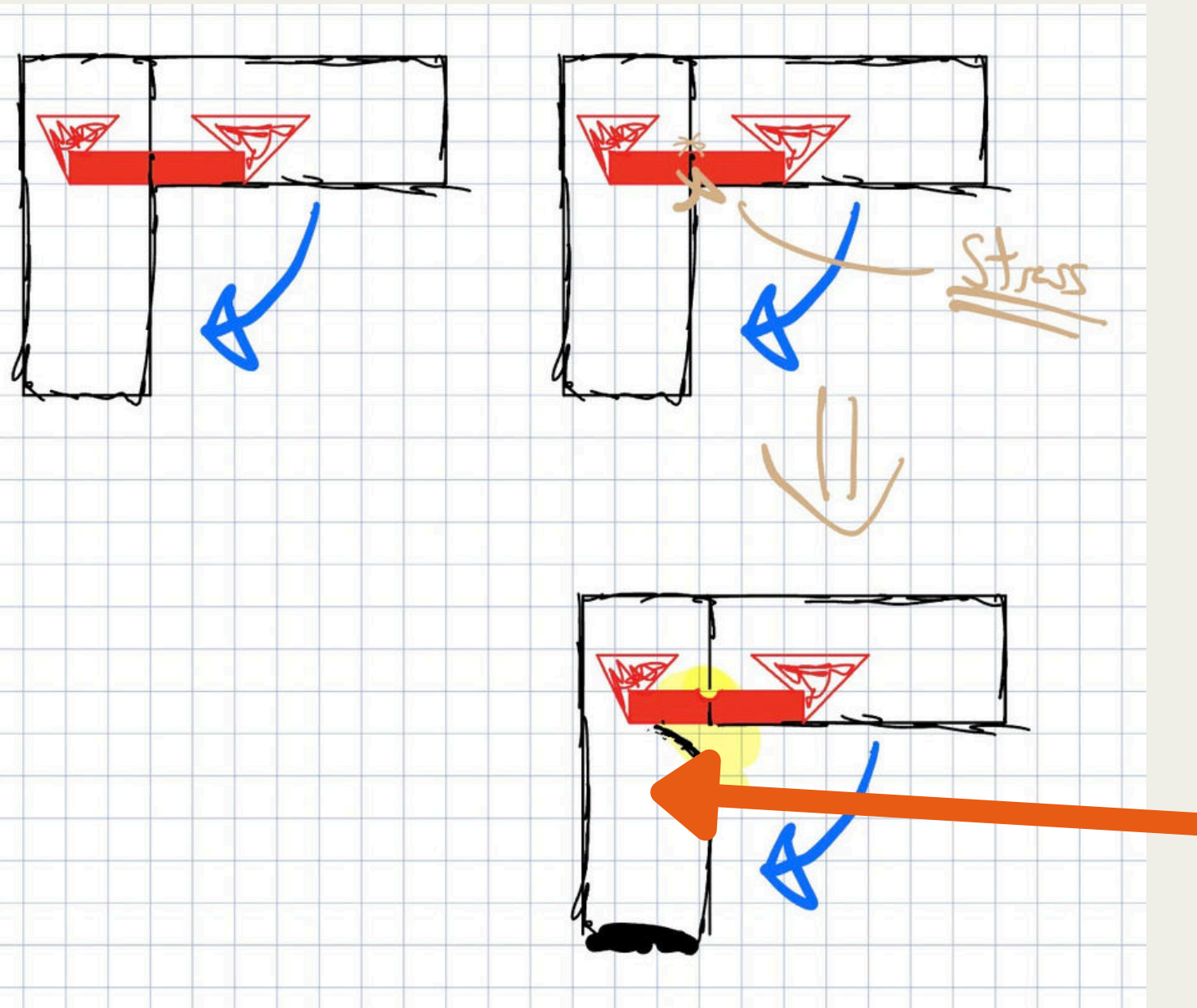


Hole in flexible
section to
complete
restraint.



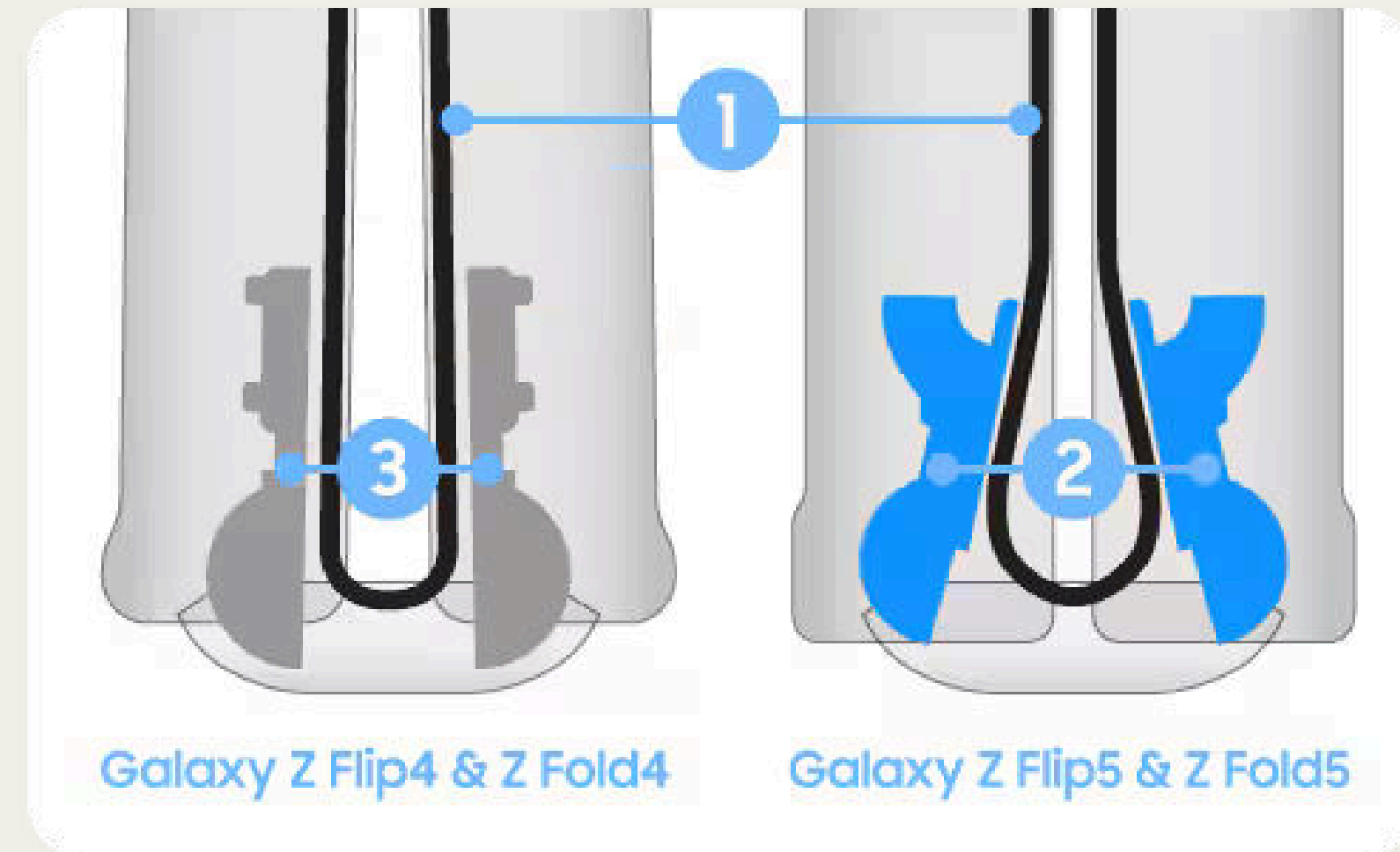
Geometry (only
hope)

Idea Generation



Modify geometry to reduce stress build up at single point

loop cut out



“Loop” cut out Inspired by a [video](#) I watched on foldable screens, which also try and prevent wear from repeated use.

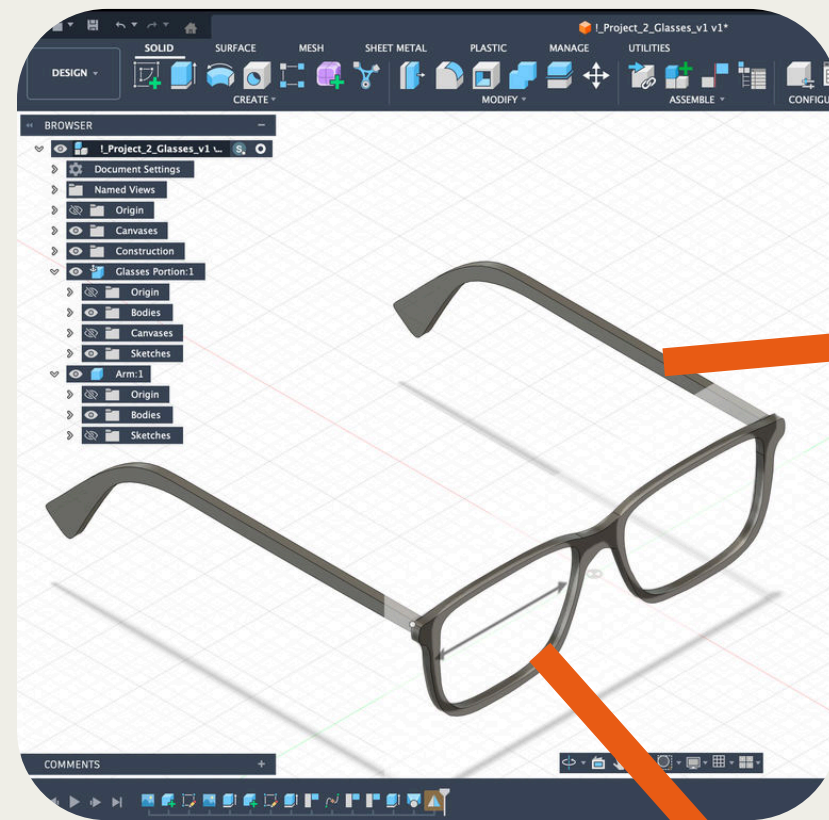
Material Selection

Rigid material was needed, **PLA** was the obvious choice due to its reliability and affordability.

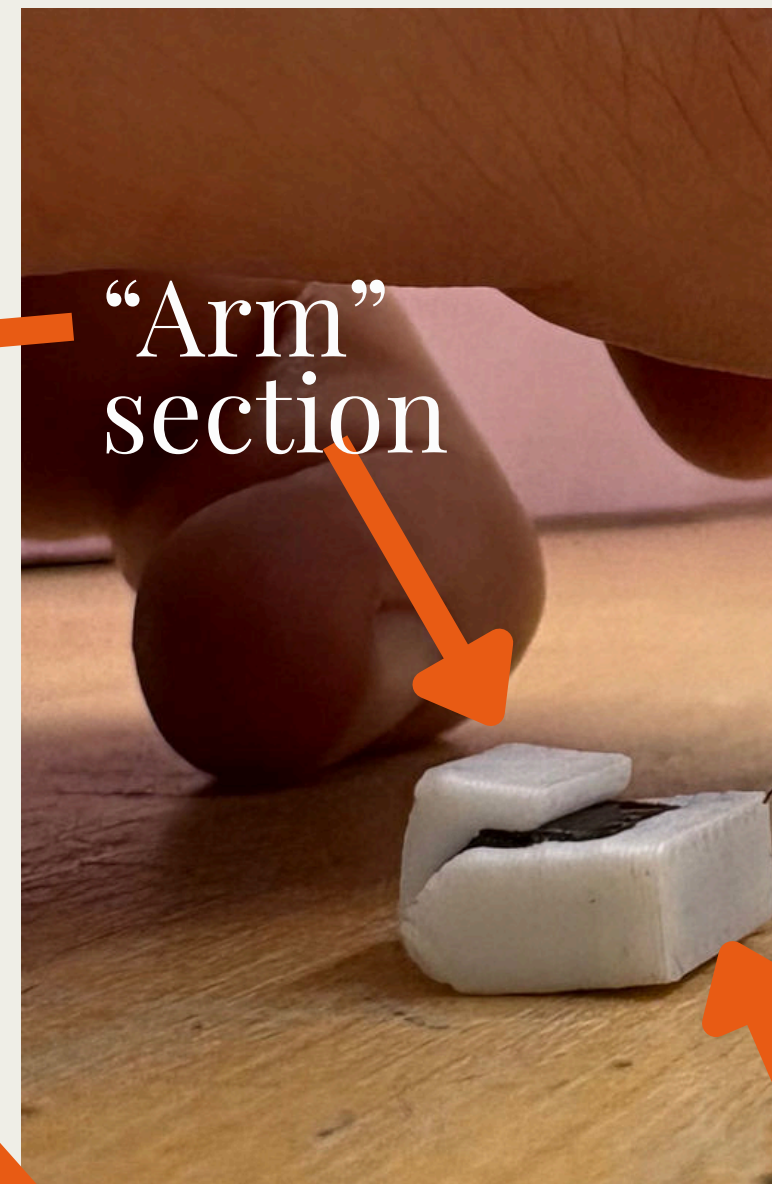
For **flexibility**, **Overture TPU** excelled in all relevant material properties, while also being available.

Property	Overture TPU (Standard)	Bambu TPU for AMS
Shore Hardness	95A	68D
Est. Flexural Modulus	~75–100 MPa	~450–600 MPa

Hinge prototypes



“frames”
side



“Arm”
section

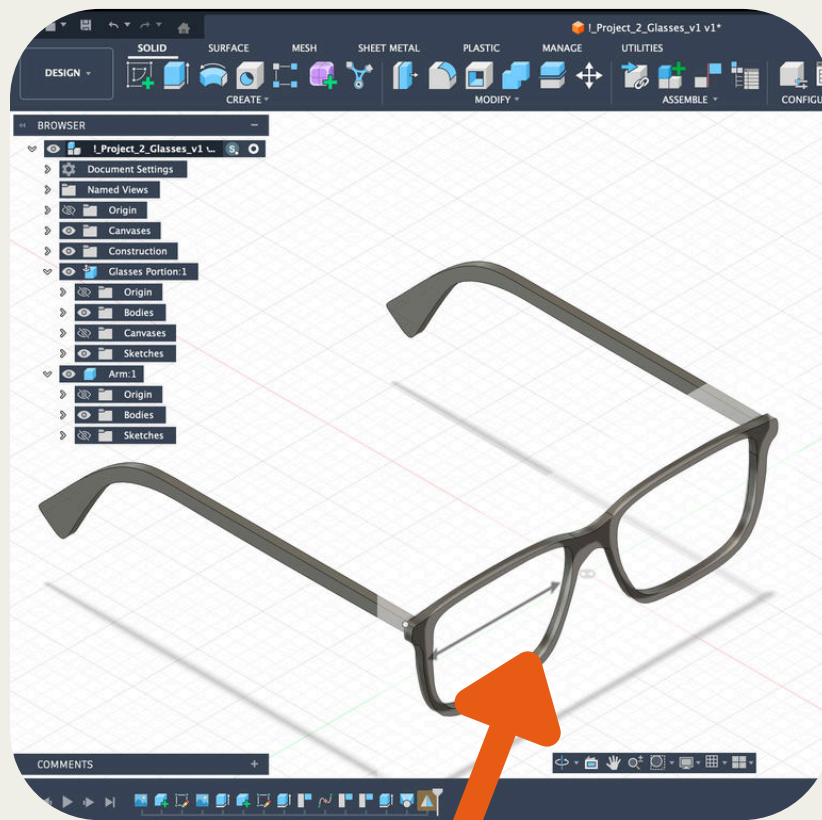
“frames” side (always the
smaller part in these prints)

Printed with the
“frames” side down.

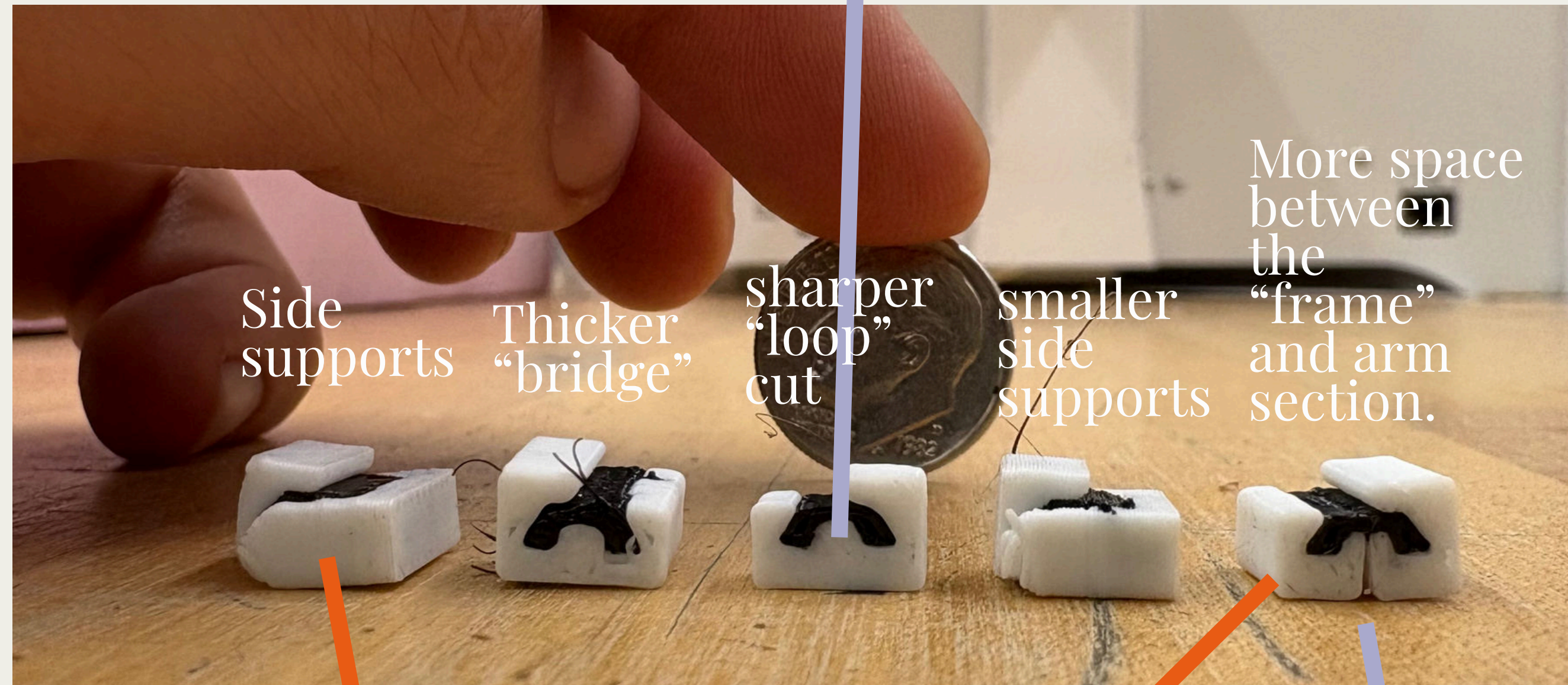
needed to **break the**
fused “frames” and
“arms”, but the hinge
functioned well.

Hinge prototypes

Note “fused” parts, from very little space between parts in CAD



“frames”
side

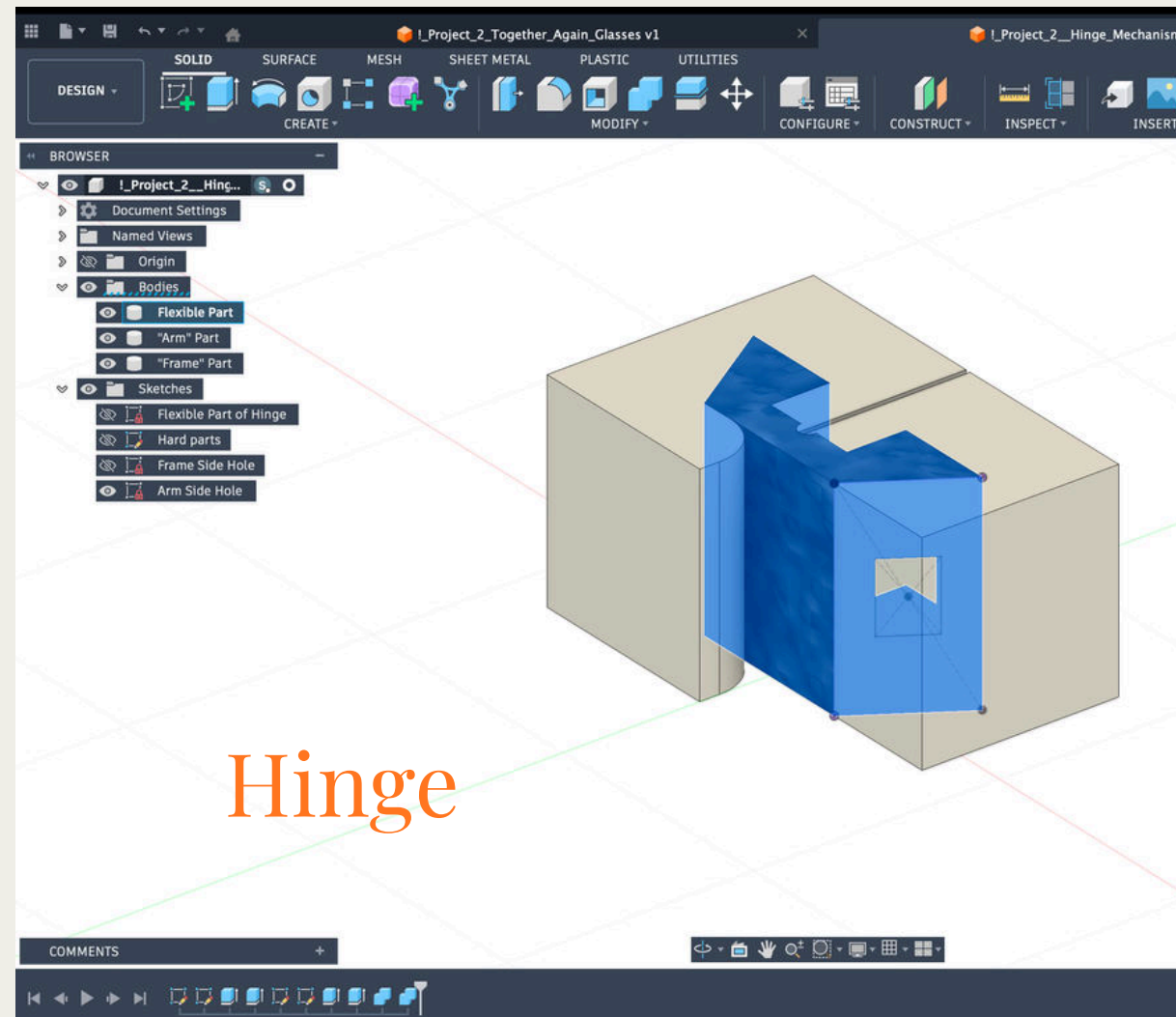


Combine “side supports”(only well functioning hinge) print orientation with “more space...” to create optimal functional hinge.

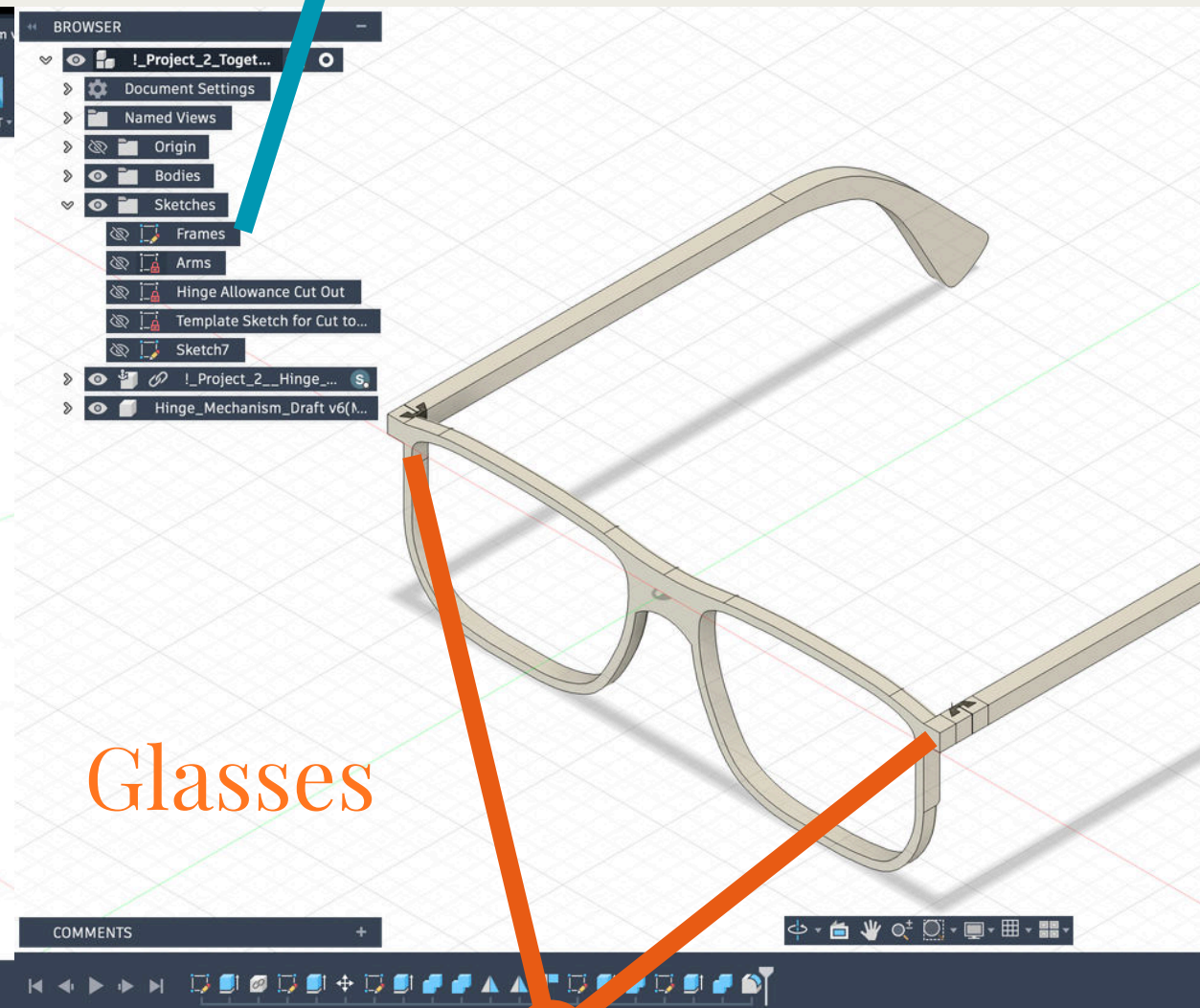
only not
fused

Final CAD

Frames sketches not fully restrained due to Spline Use



Hinge



Glasses

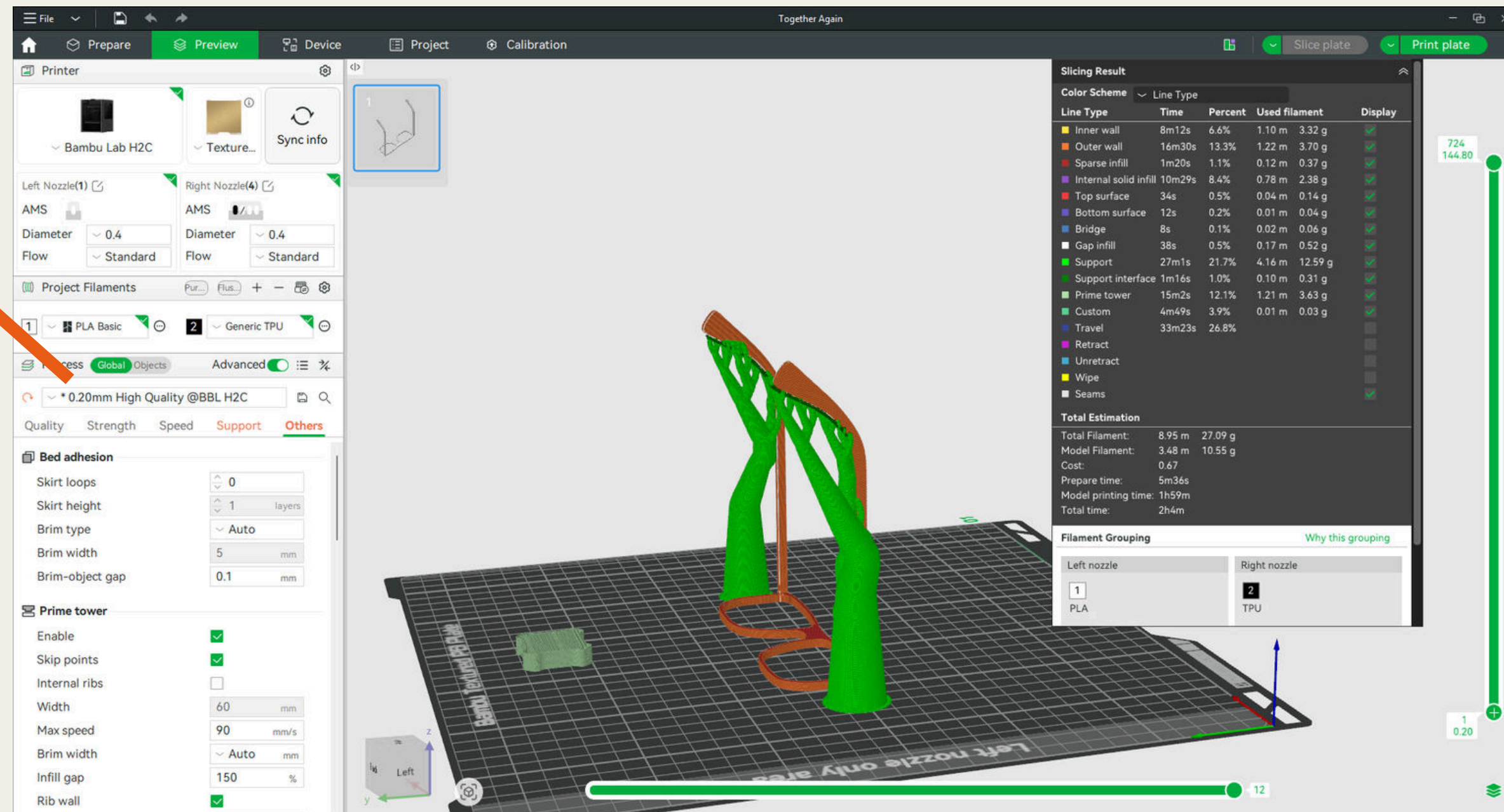
Note the offset hinges, allowing both arms to fold flat

No filets on flat printing side; minimal use elsewhere for aesthetics.

Slicer

Frames sketches not fully restrained due to Spline Use

“High quality”
print mode
used





Always Together Glasses

ME 227, Winter 2026
Sebastian Esteva



I started with lofty goals of a multi-material print, then being mid-print inserted into a print in the MarkForge Mark Two. **That did not come to pass.**

I switched to a highly custom fit pair of glasses with a great, simple yet **SMALL** hinge mechanism.

I **succeeded on the small**, and that to me feels like enough of a victory; however, I don't feel proud of this project to wear around (yet), due to its flimsy, breakable nature.

- 1) Always **double your timeline** when working with **new equipment**; the new H2Cs caused me many issues.
- 2) **Small scales** make every principle of 3d design (print orientation, slicer settings, ect) **WAY less tolerant** for mess-ups.

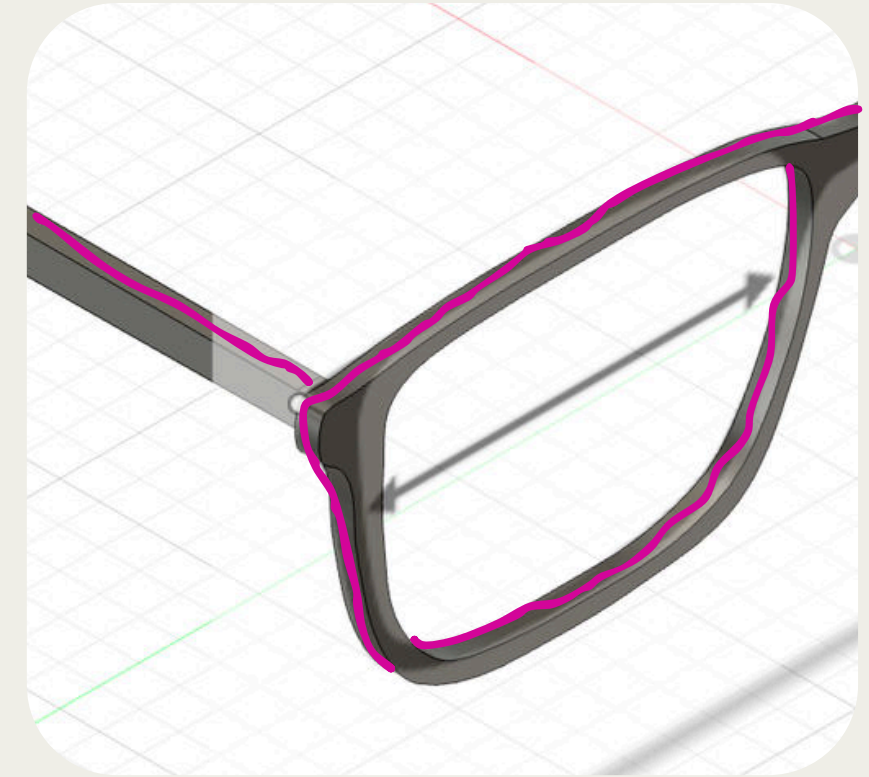
3 Learnings

- 3) When CADing at small scales, **better results** come from using **multiples of your layer thickness or nozzle size** (i.e. 2 or 4 mm).

1) Try running TPU down the center of the print for strength, hinge bonding, and aesthetics.

2 Future Improvements

2) Try the beautiful hinge in the image; I discovered it only at the project's end!



[Metal 3d printed glasses](#)